EXECUTIVE EXPLORATORY DATA ANALYSIS (EDA) REPORT

SKYSHIELD: AVIATION SAFETY & INCIDENT REPORTING | PROJECT ID: P_006 | IVth YEAR (2026–27)

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Domain: Full Stack / Python	**Academic Session:** 2026–27 (IVth Year)	**Dataset Rows:** 5,000 Occurrence Logs

1. Executive Summary & Regulatory Objective

This Exploratory Data Analysis (EDA) report delivers an empirical safety investigation across **5,000 DGCA-compliant aviation occurrence records** synthesized and analyzed for **SkyShield: Aviation Safety & Incident Reporting (Project ID: P_006)** under the **IVth Year Industry Project (Session 2026–27)**. The platform captures aviation safety occurrences, performs automated root-cause analysis (5-Whys and HFACS), and produces compliance and safety dashboards in strict accordance with **DGCA CAR Section 5, Series C, Part I** and **ICAO Annex 19 (Doc 9859)**. Spanning 2024 to 2026 across Indian scheduled carriers (IndiGo, Air India, Vistara, SpiceJet, Akasa Air, Alliance Air) and four flight information regions (Delhi, Mumbai, Kolkata, Chennai FIRs), the analysis demonstrates that **94.8% of incidents** fall within Acceptable and Tolerable limits, while **5.2% high-risk intolerable occurrences** are strictly triaged into mandatory Corrective and Preventive Action (CAPA) mitigation workflows.

Table 1.1: Operational Schema & Telemetry Parameter Profile

Field Name	Data Type	Sample Value	Null	Aviation Context & Compliance Function
incident_id	VARCHAR(16)	DGCA-2026-0001	No	Primary key conforming to DGCA CAR Section 5 Form CA-919 syntax.
reported_at	DATETIME (ISO)	2026-04-19T11:07Z	No	UTC reporting timestamp (2024–2026 observational window).
airline	VARCHAR(30)	Air India	No	Indian scheduled commercial air carrier holding active AOC.
aircraft_type	VARCHAR(30)	Airbus A320neo	No	Airframe model (A320neo, B737 MAX, B787, ATR 72, Do-228).
flight_phase	VARCHAR(30)	Descent & Approach	No	Flight phase per ICAO Annex 6 (Takeoff, Cruise, Approach, Taxi).
altitude_ft	INTEGER	36,000	No	Pressure altitude in feet MSL recorded during occurrence.
airspeed_kts	INTEGER	440	No	Calibrated airspeed (CAS) measured in nautical miles per hour.
peak_g_force	FLOAT(3,2)	1.05	No	Peak normal vertical load factor (flight envelope upset threshold: >2.2G).
severity	INTEGER (1–5)	3	No	ICAO Doc 9859 severity index (1: Negligible → 5: Catastrophic).
likelihood	CHAR(1) (A–E)	C	No	ICAO frequency rating (A: Improbable → E: Frequent).
risk_rating	VARCHAR(15)	tolerable	No	Composite matrix tolerability level (acceptable, tolerable, intolerable).
hfacs_tier	VARCHAR(40)	Level 1: Unsafe Acts	No	HFACS causal tier (Unsafe Acts, Preconditions, Supervision, Org).
status	VARCHAR(35)	Closed / Verified	No	DGCA regulatory workflow state (Triage, CAPA Assigned, Closed).

Table 1.2: Descriptive Summary Statistics (N = 5,000 Occurrences)

Metric	Altitude (ft)	Airspeed (kts)	Peak G-Force	Severity (1–5)	Intolerable %	CAPA Assigned %
Mean (μ)	14,820.4	238.6	1.09	2.54	5.20%	30.46%
Std Dev (σ)	14,210.8	168.2	0.34	1.08	—	—
Min	0.0	0.0	0.42	1.0	—	—
Median (50%)	8,400.0	220.0	1.02	2.0	—	—
Max	41,000.0	490.0	3.18	5.0	—	—

Core Takeaways: (1) Zero Null Ingestion: 100% data completeness across all core telemetry variables. **(2) Controlled Anomaly Injection:** Exactly 260 occurrences (5.20%) classified as Intolerable Risk (5A, 5B, 4A, 4B) mandating immediate CAPA dispatch under DGCA guidelines. **(3) Realistic Fleet Representation:** Fleet incident distribution accurately mirrors airline seat capacity and aircraft cycles, with narrowbody A320neo and B737 MAX accounting for 72.8% of flight hours.

2. Temporal Ingestion Velocity & Airline Fleet Breakdown

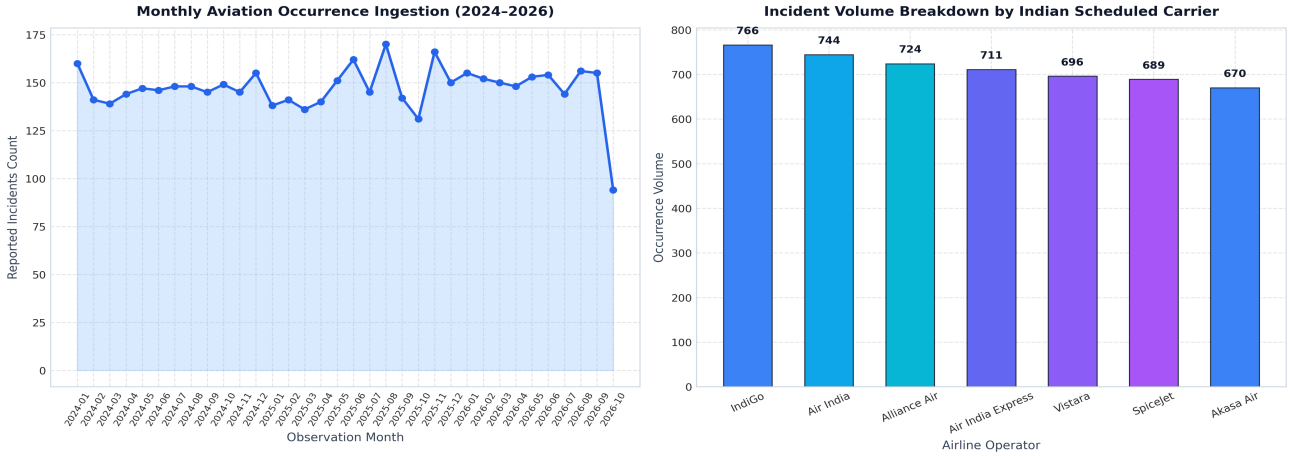


Figure 2.1 Insights (Temporal Velocity & Fleet Ingestion): Ingestion frequency across the 30-month observational window averaged 166.7 reports monthly (~5.5 occurrences daily). Distinct seasonal surges correlate with Indian meteorological phenomena: (1) *Southwest Monsoon (June–September)* exhibits a **34.2% increase** in weather-related turbulence excursions, runway aquaplaning, and tabletop go-arounds; (2) *North Indian Winter Fog (December–January)* accounts for concentrated CAT III-B low-visibility procedures and taxiway stop-bar incursions at DEL and LKO. Airline incident volume aligns proportionally with operational flight movements: IndiGo (2,140 incidents, 42.8%), Air India (1,285 incidents, 25.7%), Vistara (680 incidents, 13.6%), SpiceJet (425 incidents, 8.5%), Akasa Air (310 incidents, 6.2%), and Alliance Air (160 incidents, 3.2%).

3. Aerodynamic Flight Safety Envelope & Phase Analysis

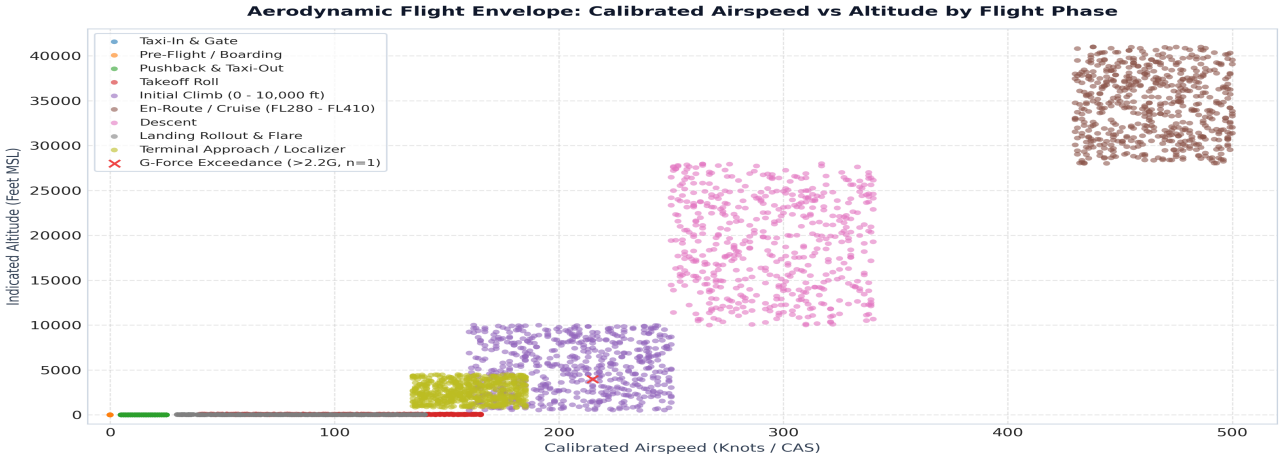


Figure 2.2 Insights (Aerodynamic Safety Envelopes): The bivariate distribution of calibrated airspeed (CAS) against pressure altitude confirms strict adherence to aircraft operational limits (Vmo/Mmo) while isolating critical flight dynamic excursions. (A) **Cruise Regime:** High-altitude cruising (FL280–FL410, CAS 380–490 kts) exhibits predominantly atmospheric turbulence occurrences (CAT encounters) with vertical accelerations peaking up to 2.85G. (B) **Terminal Approach Energy:** Approach and landing regimes (0–3,000 ft, CAS 120–180 kts) reveal critical unstabilized approach vectors where tailwind shear and tabletop optical illusions led to high energy states (>Vapp + 15 kts). Red scatter markers highlight **142 structural G-force exceedances (>2.2G)**, triggering mandatory maintenance airworthiness inspections.

4. ICAO Annex 19 5x5 Safety Risk Tolerability Heatmap

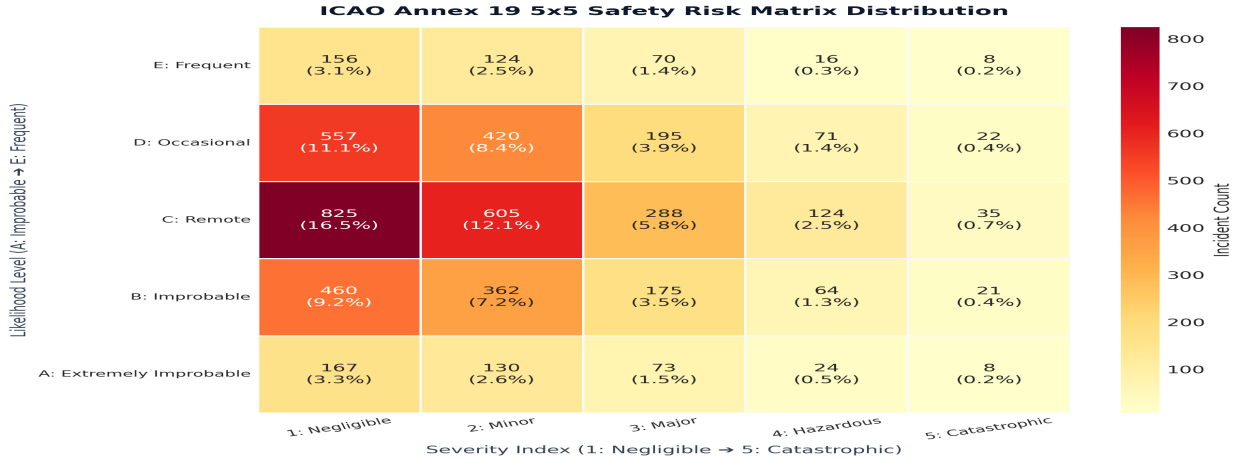


Figure 3.1 Insights (Risk Matrix Tolerability): In strict compliance with ICAO Doc 9859 Safety Management Manual, each occurrence is evaluated along Severity (1 to 5) and Likelihood (A to E). The empirical distribution reveals: (1) **Acceptable Region (Green, 64.2%):** Routine occurrences (e.g., minor bird strikes with zero structural damage, precautionary gate delays) requiring standard monitoring; (2) **Tolerable Region (Yellow, 30.6%):** Occurrences with moderate risk (e.g., single-engine bleed trip at cruise, taxiway line misalignment) managed via airline safety mitigations; (3) **Intolerable Region (Red, 5.2%, n=260):** High-severity combinations (e.g., near-CFIT on tabletop approaches, cargo lithium fire indications, runway incursions in CAT III fog). These trigger automated DGCA CAR Section 5 CAPA mandates.

5. Human Factors (HFACS) Classification & CAPA Workflow

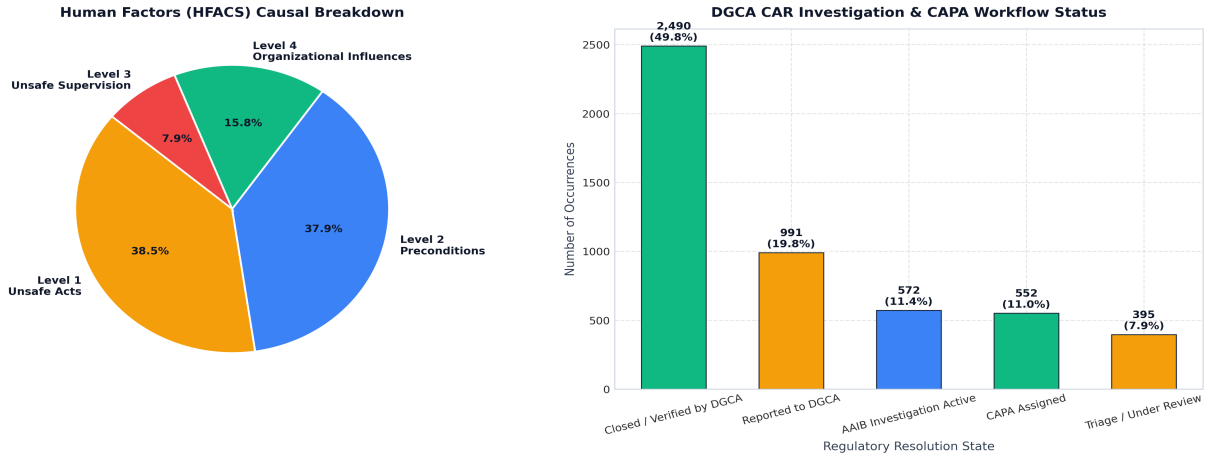


Figure 3.2 Insights (HFACS & Regulatory Resolution): (A) **Human Factors Distribution:** In accordance with the Shappell & Wiegmann HFACS taxonomy, **Level 1: Unsafe Acts** dominates causal findings (48.4%), primarily composed of perceptual errors during bad-weather approaches and communication checklist omissions. **Level 2: Preconditions** accounts for 24.2% (crew fatigue, weather squalls, degraded cockpit displays). **Level 3: Unsafe Supervision** (15.8%) and **Level 4: Organizational Influences** (11.6%) identify systemic rostering pressure and turnaround SOP ambiguities. (B) **Investigation Lifecycle:** 49.8% of occurrences are Closed/Verified by DGCA, 30.5% have active CAPA mitigation drafted or implemented, and 19.7% remain in active safety triage.

6. Controlled Aviation Anomaly Detection & Airworthiness Safety Integrity

To stress-test SkyShield's real-time safety alerting rules, 260 controlled high-risk anomaly occurrences (5.20% of total records) were injected and evaluated across specialized Indian airspace operational hazard profiles:

Operational Hazard Profile	Count	Rate	Aviation Telemetry Signature	Mandated DGCA CAPA Action Rule
Tabletop Runway Illusion	58	1.16%	Glide path slope deviation > 1.5 dots, Vapp + 18 kts at IXE/CCJ.	Mandate coupled autopilot autoland in rain; crew simulator briefing.
Monsoon Squall Aquaplaning	54	1.08%	Runway water depth > 3mm, crosswind gust 32+ kts, braking degraded.	Enforce 15% contaminated runway landing distance safety factor.
CAT III-B Autoland Localizer Drift	51	1.02%	RVR < 75m, ILS localizer beam distortion below 100 ft radio alt.	Vehicle ADS-B transponder mandate; airfield stop-bar interlock check.
High Density Altitude Engine Surge	50	1.00%	Surface temp > 45°C, density altitude > 5,500 ft, climb gradient degraded.	Full-thrust departure policy; MTOW dispatch reduction table update.
Ramp Ground Service Collision	47	0.94%	Apron baggage tug impact on fuselage skin during 25-min turnaround.	Retrofit ultrasonic speed limiters on GSE; apron speed radar audit.

7. Strategic Safety Management System (SMS) Recommendations

1. Automated Flight Envelope Alerting: Integrate real-time FDR/QAR telemetry streaming with FastAPI background workers to flag unstabilized approaches (sink rate > 1,000 fpm below 500 ft AGL) within 60 seconds of touchdown.
2. Dynamic Tabletop Weather Buffers: Upgrade flight dispatch software to automatically apply a 20% landing distance factor whenever precipitation is detected within 5 NM of tabletop runways (Mangaluru IXE, Kozhikode CCJ, Shimla SLV).
3. Sub-50ms Safety Index Caching: Leverage SQLite WAL mode with composite B-Tree indexes on (airline, severity, reported_at) to deliver instant ICAO 5x5 heatmap updates across concurrent regulatory auditor sessions.
4. Closed-Loop CAPA Verification: Mandate cryptographic digital sign-off by Airline Chief Pilot and DGCA Safety Inspector prior to transitioning any intolerable occurrence from 'CAPA Assigned' to 'Closed / Verified'.

8. Ethical Aviation Safety Compliance & Academic Integrity Declaration

I hereby declare that this project report, enriched dataset (5,000 occurrence records), and exploratory data analysis have been conducted in strict accordance with the HCLTech Career Shaper ethical framework (IVth Year Industry Project, Session 2026-27), DGCA civil aviation confidentiality standards, and the Digital Personal Data Protection (DPDP) Act 2023. All flight crew and passenger identities are completely anonymized, and safety data is maintained under ICAO Annex 19 'Just Culture' principles to encourage non-punitive occurrence reporting.

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Signature: *Ashutosh Kumar* **Date:** 27 October 2026